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Hydrogen production from steam–methanol reforming: thermodynamic analysis

Ye Lwin, Wan Ramli Wan Daud, Abu Bakar Mohamad, Zahira Yaakob*

Department of Chemical and Process Engineering, National University of Malaysia, UKM Bangi 43600, Selangor, D.E., Malaysia

Abstract

Thermodynamic equilibrium involved in the steam reforming of methanol is re-examined to cover the extended range of compounds suggested by literature to be involved in the reactions. The equilibrium concentrations are determined for different mixtures of these compounds at 1 atm and at different temperatures (360–573 K) and at different steam/methanol molar feed ratios (0–1.5), by the method of direct minimization of Gibbs free energy. The possibility of carbon formation in these conditions is determined by direct inclusion of carbon in the objective function of the minimization scheme. Results showed that the area of carbon formation region is surprisingly high. Carbon and methane formations are thermodynamically favoured and they reduce the quantity and quality of hydrogen produced. Dimethyl ether formation occurs at low temperatures and low steam/carbon feed ratios, while carbon monoxide occurs at high temperatures and low steam carbon ratios.

1. Introduction

Currently, increasing attention is being paid to the low temperature steam reforming of methanol to produce high purity hydrogen to use as fuels in fuel cells for on-board power generation for vehicles [1,2]. The favour for methanol as a chemical carrier for hydrogen is mainly due to its ready availability, high energy density, and easy storage and transportation [2–4].

The most widely used catalysts for this reaction are copper containing catalysts since copper has been found to be highly active and selective for hydrogen [5,6]. According to Refs [5–10], various types of copper catalysts have been used to give slightly different components and concentrations and hence, different mechanisms to the overall reaction:

$$CH_3OH + H_2O = CO_2 + 3H_2 \quad (1)$$

This shows the various influences of the different types and preparation methods of heterogeneous catalysts on thermodynamic equilibrium. In this study, thermodynamics of the steam–methanol reforming system is investigated to know the equilibrium compositions within the operating range of interest. From this, ideal conditions for the reaction system to maximize hydrogen production and minimize undesirable products can be determined.

Amphlett et al. [3] examined the thermodynamics of four different models to determine the effect of carbon and methane formations on steam–methanol reforming at different temperatures, pressures and feed ratios,

* Corresponding author. Tel.: +60-3-8296420; fax: +60-3-8252546.

E-mail address: zahira@vlsi.eng.ukm.my (Z. Yaakob)

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Nomenclature

a_{ji}	number of gram atoms of element j in a mole of species i	ΔG^0_{fi}	standard Gibbs free energy of formation of species i
b_j	total number of gram atoms of element j in the reaction mixture	K	total number of atomic elements
C(g)	gaseous carbon	N	total number of species in the reaction mixture
C(s)	graphite	n_i	number of moles of species i
DME	dimethyl ether	NG	total number of gaseous components
$\hat{f}_i$	fugacity of species i in the gas mixture	nG	total Gibbs free energy of the system
f^0_i	fugacity of species i at its standard state	P	total pressure of the system
G	Gibbs free energy	R	gas constant
G^0_i	Gibbs free energy of species i at its standard state	T	temperature of the system
G_i	Gibbs free energy of pure species i at operating conditions	y_i	mole fraction of species i
$\bar{G}_i$	Gibbs free energy of species i in the gas mixture	$\hat{\phi}_i$	fugacity coefficient of species i in the gas mixture

based on the widely accepted decomposition-shift mechanisms:

$$CH_3OH = CO + 2H_2 \tag{2}$$

$$CO + H_2O = CO_2 + H_2 \tag{3}$$

Thermodynamic analyses prior to them were either based on the overall reaction (1) or calculated at a fixed feed ratio [3]. In all of these investigations the method of determining equilibrium concentrations was based on stoichiometry and equilibrium constants of known reactions.

Recently, Maggio et al. [11] applied such models to make comparative study of the internal steam reforming of methane, methanol and ethanol in a molten carbonate fuel cell. But carbon formation was not considered in their calculations. Most of the previous investigators (e.g. [12]) used the principle of equilibrated gas [13] to predict the carbon formation in steam–hydrocarbon reforming. Recently Vasudeva et al. [14] estimated the carbon concentration in steam–ethanol reforming by the method of Gibbs energy minimization.

In our investigation we consider four different sets of possible compounds indicated by kinetics investigators using different copper containing catalysts. The temperature range of 360–573 K and steam/methanol molar feed ratios from zero (methanol decomposition) to 1.5 (excess steam) are used in the investigation. The lower limit of the temperature range considered is 360 K since it is about the operating temperature of a typical Proton Exchange Membrane Fuel Cell (PEMFC) to which the steam reformer is to be coupled. The pressure is kept constant at 1 atm (101.32 kPa) since previous investigations [2–4] have already shown that the higher pressures are thermodynamically not favoured by steam–methanol reforming. The method of direct Gibbs energy minimization is used to determine the equilibrium concentrations. The carbon formation is also directly estimated by describing a way of incorporating carbon concentration into the objective function of the minimization scheme.

2. Thermodynamic analysis

The total Gibbs free energy of a reacting system reaches a minimum at equilibrium. The total Gibbs function for a system is given by

$$nG = \sum_{i=1}^{N} n_i \bar{G}_i = \sum n_i G^0_i + RT \sum n_i \ \ln \frac{\hat{f}_i}{f^0_i} \tag{4}$$

For gas phase reactions, $\hat{f}_i = \hat{\phi}_i y_i P$. Although the fugacity is closed to the pressure at the condition of the calculation (1 atm), we include it for a more general case.

Since the standard state is taken as the pure ideal gas state at 1 atm, $f^0_i = 1$ atm, and since G^0_i is set equal to zero for each chemical element in its standard state, $\Delta G^0_i = \Delta G^0_{fi}$ for each component. Substituting these into (4):

$$nG(n_i's,T,P) = \Sigma n_i \Delta G_{fi}^0 + \Sigma n_i RT \ln P + \Sigma n_i RT \ln y_i + \Sigma n_i RT \ln \hat{\phi}_i \quad (5)$$

The problem now is to find the set of $n_i's$ which minimizes nG at constant T and P, subject to the constraints of elemental balances:

$$\sum_{i=1}^{N} n_i a_{ji} = b_j, \quad j = 1,\ldots,K \quad (6)$$

The objective function (5) is minimized using the IMSL Math library routine DLCONF or LCONG [15]. The routines are based on Powell's TOLMIN [16], which solves linearly constrained optimization problems by the Sequential Quadratic Programming (SQP) method. The routine DLCONF uses a finite-difference method to estimate the gradient of the objective function by high precision arithmetic for accuracy, whereas the routine LCONG uses the analytical gradient provided by the user. The gradient of the objective function can be readily worked out as:

$$\frac{\partial(nG)}{\partial n_i} = \Delta G_{fi}^0 + RT \ln P + RT \ln y_i + RT \ln \hat{\phi}_i, \quad i = 1,\ldots,N \quad (7)$$

The problems are also solved by the Lagrange multiplier method using IMSL's nonlinear equations solver DNEQNF routine. The results are found to be the same.

The Redlich–Kwong equation of state is used to calculate the fugacity coefficient of each component in the gas mixture [17]. The critical properties and ideal gas standard Gibbs free energy of formation as a function of temperature for each component are obtained from HYSYS's pure component library database [18]. Since 1 atm pressure is used, the ideal gas behaviour can also be assumed and consequently $\hat{\phi}_i's$ are unity and the last terms of (5) and (7) will diminish.

The total Gibbs function (4) applies to any homogeneous phases at reaction equilibrium. But the carbon formation in gas phase reactions can be estimated by exploiting the phase equilibrium existed between solid carbon and carbon vapour in the gas phase [19,20]:

$$\bar{G}_{C(g)} = \bar{G}_{C(s)} \quad (8)$$

If the carbon formed is considered to be pure graphite form, then

$$\bar{G}_{C(s)} = G_{C(s)} \cong \Delta G_{f,C(s)}^0 = 0 \quad (9)$$

From (8) and (9) it follows that

$$(n\bar{G})_{C(g)} = (n\bar{G})_{C(s)} \quad (10)$$

Then the condition of equilibrium, replacing the total Gibbs energy of carbon vapour with that of solid carbon, becomes

$$\min(nG) = \sum_{i=1}^{NG} n_i \bar{G}_i + (n\bar{G})_{C(s)} \quad (11)$$

where NG is the number of substances which are present only in the gas phase while carbon is also present as solid.

Substituting (5) for gas phase components and (9) for carbon into (11), the objective function when carbon formation is considered becomes

$$nG(n_i's,T,P) = \sum_{i=1}^{NG} n_i(\Delta G_f^0 + RT \ln P + RT \ln y_i + RT \ln \hat{\phi}_i) + (n\Delta G_f^0)_{C(s)} \quad (12)$$

The constraints are the same as (6) with $N = NG + 1$. For the amount of carbon vapour, it can be considered to be non-existent since carbon has extremely small vapour pressure at the temperatures being considered. However, it is obvious from (12) that the total Gibbs free energy still depends on the solid carbon formed.

Although it can be seen from (9) that the last term of (12) is zero, it is necessary to incorporate this term in the objective function for numerical stability in solving this problem. In their article on thermodynamic analysis of steam–ethanol reforming, Vasudeva et al. [14] described that the total Gibbs free energy can be considered to be independent of carbon based on its negligible vapour pressure and Eq. (9), and only included it in the elemental constraints. Since this can lead to numerical instabilities in solving the problem possibly due to the difference in the number of unknowns between the objective function and constraints, we include the last term in the objective function.

In the present investigation, we consider four different cases of compounds indicated in the Refs [5–10] to be involved in copper-catalyzed steam reforming of methanol. Some investigators reported only the primary compounds of steam–methanol reforming: CH_3OH, H_2O, CO, CO_2 and H_2 [6,7]. This follows from the widely accepted decomposition-shift mechanism (2) and (3). Also reported together with the primary compounds are higher molecular weight compounds formaldehyde (HCHO) and methyl formate ($HCOOCH_3$) with a different mechanism [5]. Takahashi et al. [8] and Jiang et al. [9] showed that the CO may not be involved in the reactions of steam–methanol reforming and formulated a mechanism involving $HCOOCH_3$ and formic acid (HCOOH). Trace amounts of CH_4 in steam–methanol reforming

Table 1
Equilibrium moles (per mole of methanol fed) at 500 K and 1 atm with steam/carbon feed ratio 1.0

Case	*nG*, kJ	CH_3OH	H_2O	CO	CO_2	H_2	DME	CH_4	C
1	−405.503	0.0003	0.1364	0.1354	0.8635	2.8620	Absent	–	–
3	−445.858	Absent	1.9984	Absent	0.0003	1.0006	Absent	–	0.9988
2	−461.084	Absent	1.4866	Absent	0.2562	0.0266	Absent	0.7429	–
4	−461.085	Absent	1.4994	Absent	0.2498	0.0276	Absent	0.7360	0.0133

was reported by [6] and dimethyl ether (CH_3OCH_3) and CH_4 were reported in methanol decomposition by [10].

According to these findings, we classify our four cases as follows, similar to the classification by Amphlett et al. [3] regarding the species which can exist in equilibrium.

Case 1—CH_3OH, H_2O, CO, CO_2, H_2, $HCOOCH_3$, CH_3OCH_3, HCHO, and HCOOH
Case 2—Case 1 plus CH_4
Case 3—Case 1 plus carbon
Case 4—Case 1 plus CH_4 and carbon

Case 1 represents the complete set of compounds involved in the gas phase reactions most researchers reported excluding trace amounts of CH_4, if present. CH_4 formation is considered in Case 2 and the carbon formation is considered in Case 3, respectively, along with the compounds in Case 1. In Case 4 both carbon and methane formations are considered with Case 1 compounds.

essentially in agreement with that calculated by Amphlett et al. [3]. By comparing the equilibrium concentrations of Cases 1 and 2, it becomes clear that the carbon formations in Cases 3 and 4 are due to the reactions:

$$CO + H_2 = C + H_2O \tag{13}$$

$$CO_2 + 2H_2 = C + 2H_2O \tag{14}$$

$$CH_3OCH_3 = 2C + H_2O + 2H_2 \tag{15}$$

$$CH_4 = C + 2H_2 \tag{16}$$

Reactions (13)–(15) are thermodynamically favourable because of large Gibbs energy decreases especially at low temperatures since they are exothermic. Reaction (16) is thermodynamically unlikely at low temperatures and its contribution to carbon formation is very small.

3. Results and discussion

The results show that the equilibrium concentrations of $HCOOCH_3$, HCHO, and HCOOH are zero for all cases. This means that even if these components are involved in the reaction mechanisms they are merely intermediates. From the Gibbs energy values results, it is found that the decomposition of methanol is thermodynamically less favoured than steam reforming for all cases. Also the methanol conversion and selectivity for hydrogen are higher in steam reforming than in methanol decomposition. Typical Gibbs energy values in descending order for equilibrium of steam–methanol reforming at 500 K and 1 atm with a steam/methanol molar feed ratio of 1.0 for the four cases are shown in Table 1.

Fig. 1 shows the range of conditions under which carbon will form in the system of Cases 3 and 4. The curves were plotted by determining the points corresponding to the first disappearance of carbon as the temperature is increased for a fixed feed ratio. It is found that the area of carbon formation region is

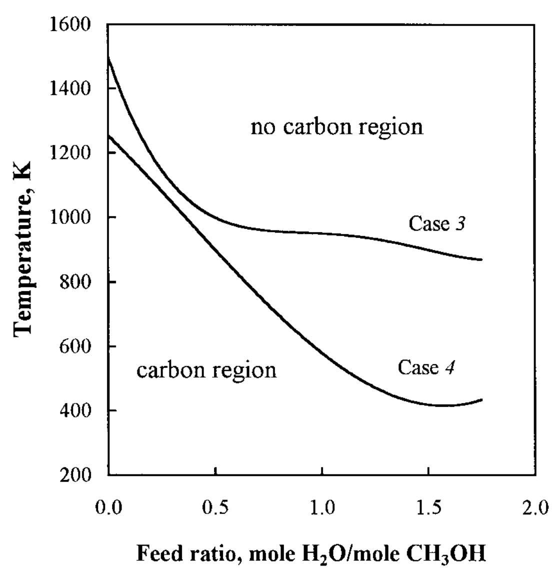

Fig. 1. Limiting conditions (temperature and feed ratio) for carbon formation at 1 atm in Cases 3 and 4.

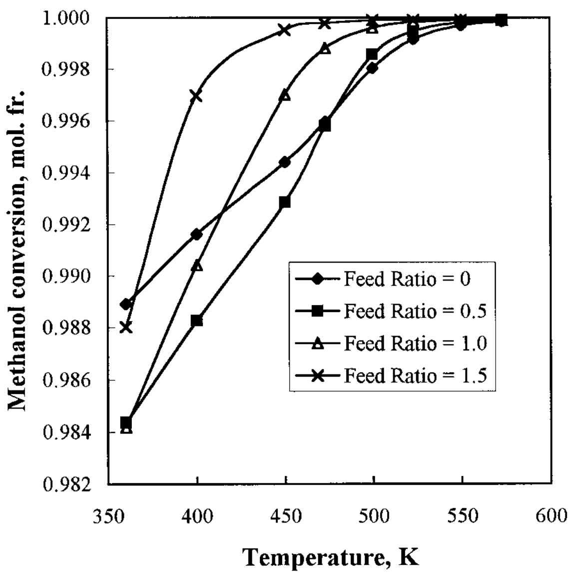

Fig. 2. Equilibrium conversion of methanol as a function of temperature and feed ratio at 1 atm for Case 1.

From Fig. 1 it is found that carbon will form in the entire operating range considered (360–573 K and a feed ratio of 0–1.5). In practice, however, carbon formation was not detected by most kinetic investigators. This may be due to several reasons including rate control rather than equilibrium control of the process so that ultimate equilibrium is never reached, surface structure of the catalysts, and type of hydrocarbon feed [13]. A similar reason as rate controlling can be given to the absence of methane from reaction products in practice. This means that the catalyst has some influence on the selectivity. Also the carbon that would actually deposit on the catalyst may have a different form with lower Gibbs free energy of formation than the graphite on which thermodynamic calculations were based [13,21]. The large differences of carbon and methane concentrations between calculated and observed ones might also be due to the limited number of species taken into account in the calculation.

By comparing the equilibrium concentration of Case 1, it is clear that the methane formation in Case 2 is due to the reaction:

$$CO + 3H_2 = CH_4 + H_2O \tag{17}$$

The conversion of methanol at different temperatures and feed ratios for Case 1 is shown in Fig. 2. In all cases methanol conversion was found to be almost 100% at all operating conditions. Thus, thermodynamic analysis encourages development of better catalysts with higher H_2 selectivity and low temperature activity for steam–methanol reforming. The permissible methanol content in the anode feed gas for fuel cells is specified as 5000 ppm for a reversible performance loss [22]. This is equivalent to 99.5% conversion of methanol. This conversion can be achieved by operating the steam–methanol reforming at the temperatures above 380 K with a steam/methanol feed ratio of 1.5, as can be seen from Fig. 2.

Fig. 3 shows the equilibrium concentrations of CO and dimethyl ether (DME) for Case 1. It can be seen that the formation of CO, which is poisonous for platinum electrodes of the fuel cells, can be minimized by operating the reforming at low temperatures and high steam/methanol ratios. In Case 3, CO is absent for all the operating conditions considered. This is due to the formation of carbon from CO according to (13). In Cases 2 and 4, CO is virtually absent from steam reforming products due to the reactions (13) and (17) which are thermodynamically highly favourable.

By referring to Fig. 3 for DME formation at equilibrium for Case 1, it is seen that the DME concentration increases with decreasing temperatures and steam/methanol feed ratios. No DME is found in Cases 3 and 4, indicating that all DME formed is decomposed to form carbon according to (15). DME is a by-product of steam–methanol reforming and can be formed by dehydration of methanol.

$$CH_3OH = \tfrac{1}{2}CH_3OCH_3 + \tfrac{1}{2}H_2O \tag{18}$$

Reaction (18) is thermodynamically less likely at the temperatures considered when initial water is present. This explains why DME is absent from steam reform-

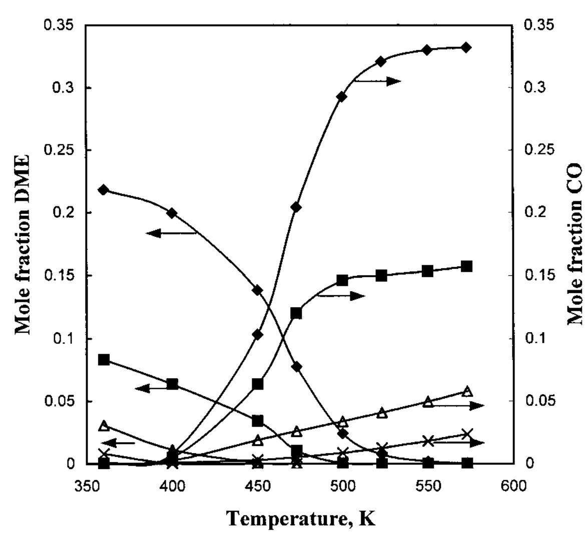

Fig. 3. Equilibrium concentrations of CO and DME as functions of temperature and feed ratio at 1 atm for Case 1 (the legends are as for Fig. 2).

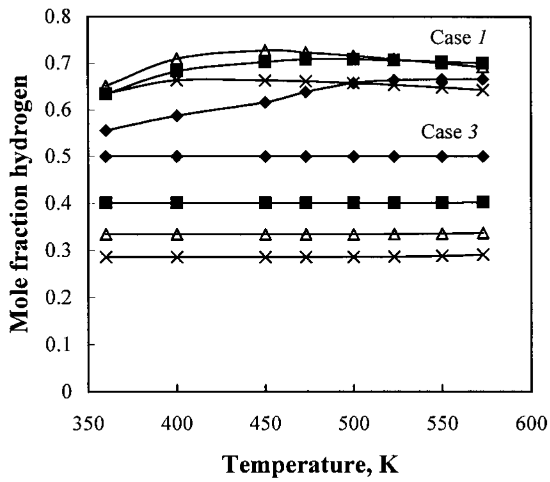

Fig. 4. Equilibrium concentration of hydrogen as a function of temperature and feed ratio at 1 atm for Cases 1 and 3 (the legends are as for Fig. 2).

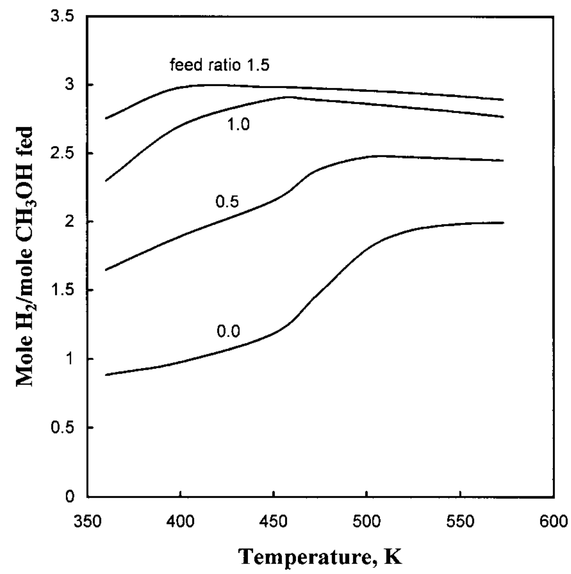

Fig. 5. Hydrogen yield (mole H_2/mole CH_3OH in feed) at equilibrium at 1 atm as a function of temperature and feed ratio for Case 1.

ing products in practice and can only be found in methanol decomposition products. Since its presence lowers the H_2 selectivity, DME should be prevented from formation. It can be seen from Fig. 3 that DME formation does not occur at practical operating conditions of temperatures higher than 450 K and steam–methanol feed ratios above 1.0.

The results show that the formation of DME can be avoided and CO can be controlled below 100 ppm by operating the steam reforming at temperatures of 400–460 K and feed ratios greater than 1.5.

The results and Fig. 4 show the effect of carbon and methane formations on hydrogen production in comparison with Case 1. In Cases 2 and 4 where CH_4 formation is considered, H_2 is nearly absent from equilibrium products. It is clear that the presence of methane and carbon reduces the quantity and quality of hydrogen produced. The role of steam as a reforming agent disappears and the steam feed remains unreacted if carbon and methane formations are considered. The quality of hydrogen is reduced not only by methane but also by steam and CO_2 which are increased in Cases 3 and 4. In these cases, in addition to the steam in feed, more steam is produced by reactions (13)–(15) and (17), while additional CO_2 is produced by water–gas shift reaction (3).

From Fig. 4, it is seen that the curve for feed ratio 1.5 lies below the curves for feed ratios 0.5 and 1.0. This is due to the presence of much higher amounts of water at equilibrium for the excess water feed ratio, which reduces the mole fraction of hydrogen, but not necessarily its quantity. This is indicated in Fig. 5 which shows the highest quantity of hydrogen is produced for excess water feed ratio at all temperatures and amounts to nearly 3 moles H_2 per mole of methanol which is the stoichiometric yield indicated in the overall reaction (1). Both Figs. 4 and 5 show that for any feed ratio in Case 1, the amount of hydrogen in the products shows a maximum at $400 \leq T \leq 500$ K. This is essentially in agreement with the works of Amphlett et al. [3] and Agaras et al. [23]. At higher temperatures the amount of hydrogen decreases again as the water–gas shift reaction (3) becomes less favoured and consequently, the amount of unreacted water increases.

4. Conclusions

1. If carbon and methane formations are not considered, the thermodynamic optimum condition for hydrogen production occurs at 1 atm pressure, 400 K and a steam/methanol feed ratio of 1.5. Under this condition the equilibrium concentration of CO is less than 1000 ppm and that of DME is less than 300 ppm, with a hydrogen yield of 2.97 moles per mole of methanol and methanol conversion of 99.7%.
2. Although the concentrations of CO and DME can be further reduced at feed ratios greater than 1.5, H_2 yield cannot be raised above the theoretical limit of 3 moles per mole of methanol, and the H_2 mole fraction will be reduced further by the presence of higher amounts of unreacted water. Kinetic con-

siderations should also be made when higher feed ratios are to be used.

3. It has been known that copper containing catalysts can prevent carbon and methane formations in steam–methanol reforming and can open up the mechanisms leading to Case 1 in the temperature range 443–573 K. But the catalysts may still need to be modified to get high activity and H_2 selectivity at lower temperatures.

5. Glossary

DLCONF Double precision version of the subroutine LCONF (linearly constrained minimization of general objective functions with finite difference gradient).

DNEQNF Double precision version of the subroutine NEQNF (nonlinear equations solver with finite difference Jacobian).

IMSL International Mathematical and Statistical Library.

LCONG Linearly constrained minimization of general objective function with analytic gradient.

TOLMIN A tolerant algorithm for linearly constrained optimization calculations.

Acknowledgements

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